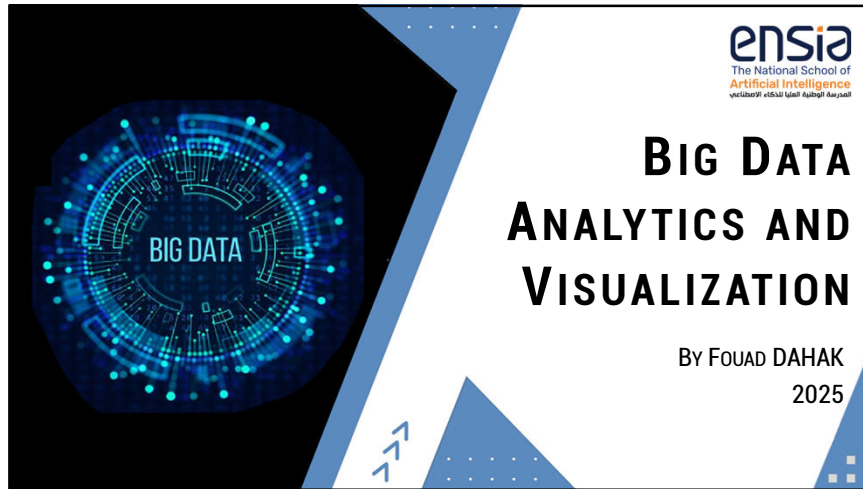




As of 2023, humans generate 3.5 quintillion bytes of data daily when searching, posting, or browsing the web

Worldwide Big Data market revenues for software and services are projected to increase from **\$42 billion** in **2018** to **\$103 billion** in **2027**, attaining a CAGR of **10.48%**.

The Big Data analytics industry will be worth **\$650 billion** by **2029**.

Career opportunities in data science will increase by **28%** by **2026**.
That equates to **11.5 million** new employment opportunities

Hadoop will hit **\$842.25 billion** in market value by **2030**

Course Objectives

- Understand the fundamentals of big data**
Define and explain the characteristics of big data and its role in artificial intelligence.
- Master big data technologies and architectures**
Explore the big data ecosystem, including Hadoop, Spark, NoSQL databases, and cloud computing solutions.
- Develop hands-on experience with big data tools**
Utilize platforms such as Apache Spark, Kafka, MongoDB, and cloud-based big data solutions.
- Apply data processing techniques**
Implement distributed computing frameworks, ETL pipelines, and real-time data streaming.
- Analyze and visualize big data**
Use advanced analytics, machine learning models, and visualization tools like Power BI and Tableau.

CONTENT

- 01 Understanding Big Data**
- 02 Hadoop Distributed File System (HDFS)**
- 03 MapReduce Programming**
- 04 Introduction to Apache Spark**
- 05 Spark SQL and Data Frames**
- 06 Data Ingestion and ETL Processes**
- 07 NoSQL Databases and Big Data**
- 08 Data Visualization and Reporting**

Big Data Characteristics: The 7Vs

→ Vs Framework

Validity

Data should be clean, accurate, reliable, valid, and useful

Volume (Laney, 2001)

Refers to the massive amount of data generated daily.

Velocity (Laney, 2001)

Describes the speed at which data is generated and processed.

Variety (Laney, 2001)

Indicates the diversity of data types (structured, semi-structured, unstructured).

Veracity (IBM, 2012)

Focuses on the quality and trustworthiness of data.

Value (Oracle, 2013)

Highlights the importance of deriving actionable insights from data.

Visualization (MS, 2015+)

Stresses the need for effective tools to represent complex data in understandable formats.

Variability (2015)

Refers to the inconsistency and dynamic nature of data patterns over time.

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Big Data Characteristics: The 7Vs

→ Characteristic 1: Volume

DEFINITION

Volume refers to the massive size of datasets generated every second across the globe.

CHALLENGES

- ❖ Traditional storage systems cannot scale to store such vast amounts of data.
- ❖ Data redundancy and ensuring data availability become complex.

Platform/Service	Data Volume in 60 Seconds
Google	973,000 Logins
Facebook	18 Million Text Messages
YouTube	4.3 Million Videos Viewed
Netflix	266,000 Hours Watched
Search Engines	3.7 Million Search Queries
App Store	375,000 Apps Downloaded
Instagram	174,000 Scrolling Instagrams
Twitter	481,000 Tweets Sent
Tinder	1.1 Million Swipes
Amazon Echo	67 Voice-First Devices Shipped
Twitch	936,073 Views
WhatsApp	25,000 GIFs Sent via Messenger
Snapchat	2.4 Million Snaps Created
Spotify	5862,823 Songs Played

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Big Data Characteristics: The 7Vs

→ Characteristic 2: Velocity

DEFINITION

Velocity refers to the speed at which data is generated, collected, and processed in real-time or near real-time.

CHALLENGES

- ❖ Processing and analyzing data streams in real-time.
- ❖ Ensuring low-latency responses for critical systems (e.g., fraud detection).

The New York Stock Exchange captures **1 TB OF TRADE INFORMATION** during each trading session.

Modern cars have close to **100 SENSORS** that monitor items such as fuel level and tire pressure.

By 2016, it is projected there will be **18.9 BILLION NETWORK CONNECTIONS** — almost 2.5 connections per person on earth.

Velocity
ANALYSIS OF STREAMING DATA

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Big Data Characteristics: The 7Vs

→ Characteristic 3: Variety

DEFINITION

Variety refers to the diversity of data formats and sources, including structured, semi-structured, and unstructured data.

CHALLENGES

- ❖ Integrating multiple data types from disparate sources.
- ❖ Processing unstructured data effectively

As of 2011, the global size of data in healthcare was estimated to be **150 EXABYTES** (100 BILLION GBs).

By 2014, it's anticipated there will be **420 MILLION WEARABLE, WIRELESS HEALTH MONITORS**.

Variety
DIFFERENT FORMS OF DATA

4 BILLION+ HOURS OF VIDEO are watched on YouTube each month.

30 BILLION PIECES OF CONTENT are shared on Facebook every month.

400 MILLION TWEETS are sent per day by about 200 million monthly active users.

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Big Data Characteristics: The 7Vs

Characteristic 4: Veracity

Lesson 01 17

DEFINITION

Veracity refers to the quality, accuracy, and trustworthiness of data.

CHALLENGES

- ❖ Poor decision-making due to unreliable insights.
- ❖ Increased complexity in cleaning and preprocessing data

1 IN 3 BUSINESS LEADERS don't trust the information they use to make decisions

27% OF RESPONDENTS in one survey were unsure of how much of their data was inaccurate

Poor data quality costs the US economy around \$3.1 TRILLION A YEAR

Veracity
UNCERTAINTY OF DATA

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Big Data Characteristics: The 7Vs

Characteristic 5: Value

Lesson 01 18

DEFINITION

Value refers to the meaningful insights and actionable benefits derived from analyzing big data. Mechanism to bring the correct meaning out of the data.

CHALLENGES

- ❖ Identifying relevant data from large datasets.
- ❖ Lack of expertise in big data analysis (data science and engineering skills).
- ❖ High costs of data infrastructure and tools.

By 2029, the value of the big data analytics market is expected to reach over \$655 billion that is double the value of today's (2023) market value!

Year	Value (Billions US dollars)
2021	240.56
2022*	271.83
2023*	308.26
2024*	349.56
2025*	396.4
2026*	449.32
2027*	509.76
2028*	578.06
2029*	655.53

[Source: Statista.com, 2023]

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Big Data Characteristics: The 7Vs

Characteristic 6: Variability

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DEFINITION

Variability refers to the inconsistency and unpredictability in data patterns over time.

CHALLENGES

- ❖ Difficulty in designing systems to adapt to sudden spikes or changes in data trends.
- ❖ Complex algorithms needed to detect and adjust to evolving patterns.

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Big Data Characteristics: The 7Vs

Characteristic 7: Visualization

Lesson 01 20

DEFINITION

Visualization refers to the graphical representation of data to make insights easier to interpret and communicate.

CHALLENGES

- ❖ Representing high-volume and multi-dimensional data clearly.
- ❖ Balancing detail with simplicity to avoid information overload.

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Big Data Characteristics: The 7Vs

Is It Big Data or Not?

Does the dataset exhibit one or more Vs?

Determining whether a dataset qualifies as big data requires evaluating it against the 7Vs framework. A dataset does not need to exhibit all seven characteristics simultaneously; even one significant attribute can place it in the big data category.

Is it beyond the capacity of traditional RDBMS to handle?

Does it require specialized tools for storage, processing, or analysis?

Big Data

technology database
structured data
unstructured data
in memory analytics
software
enterprise tools
analysis
loosely structured data
analytics
processing volume

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Big Data Characteristics: The 7Vs

Is It Big Data or Not?

Example	Volume	Velocity	Variety	Veracity	Value	Variability	Visualization
Social Media Analytics	✓	✓	✓	✓	✓	✓	✓
IoT Sensor Data	✓	✓	✓		✓	✓	✓
E-Commerce Transactions	✓	✓		✓	✓	✓	✓
Healthcare Genomic Data	✓		✓	✓	✓	✓	✓
Real-time Stock Market Analysis	✓	✓		✓	✓	✓	
Small Business Accounting Records	✓			✓	✓		
Personal Photo Collection	✓			✓			✓
Local Library Catalog	✓		✓	✓			

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Lesson N° 2

APPLICATIONS OF BIG DATA

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Applications of Big Data

Types of Big Data Analytics



Descriptive
Explains what happened
Based on Live Data, Tells what's happening in real time
Accurate & Handy for Operations management
Easy to Visualize



Diagnostic
Explains why it happened
Automated RCA – Root Cause Analysis
Explains “why” things are happening
Helps trouble shoot issues



Predictive
Forecasts what might happen
Tells What's likely to happen?
Based on historical data, and assumes a static business plans/models
Helps Business decisions to be automated using algorithms.



Prescriptive
Recommends an action based on the forecast
Based on current data analytics, predefined future plans, goals, and objectives
Advanced algorithms to test potential outcomes of each decision and recommends the best course of action

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Applications of Big Data

Real-life examples of big data

Lesson 02

25

Social Media



- User interactions, posts, likes, shares, comments.
- Facebook processes over 4 petabytes of data daily.
- Analyze user behavior, sentiment analysis, targeted advertising.

E-Commerce



- Transactions, user browsing patterns, product reviews
- Millions of transactions processed daily
- Predict customer preferences, optimize inventory, improve recommendations

IoT Devices



- Sensor readings (temperature, heart rate, etc.)
- Billions of real-time data points generated daily
- Enhance automation (smart homes), personalized health monitoring

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Applications of Big Data

Real-life examples of big data

Lesson 02

26

Healthcare



- Electronic health records, medical imaging, genome sequencing.
- Petabytes of data from hospitals and research centers.
- Predict disease outbreaks, improve diagnostics, enable precision medicine.

Financial Services



- Stock trades, credit card transactions, loan applications.
- Thousands of transactions per second globally.
- Detect fraudulent activity, analyze market trends, optimize trading algorithms

Agriculture



- Use satellite imagery and machine learning algorithms to identify crop stress, nutrient deficiencies, and other issues that can affect crop yields
- Allows farmers to take targeted actions to address these issues and improve their yields

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Applications of Big Data

Examples from real life

Lesson 02

27

by Chris Kanaracus
U.S. Correspondent

Memphis Cops Use Analytics to Fight Crime

News
21 Jul 2010 • 3 mins

It can sometimes be hard to pinpoint the return on investment for big data technology, but police officials in Memphis, Tennessee, have seen major results from using predictive analytics to fight crime.



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Summary

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BIG DATA ARCHITECT Designs and builds data pipelines, storage solutions, and large-scale data processing systems \$130,000 - \$200,000 PER YEAR	DATA ENGINEER Builds and maintains data pipelines, ETL processes, and big data platforms \$110,000 - \$170,000 PER YEAR	BIG DATA CONSULTANT Advises companies on big data strategies, architectures, and implementations \$120,000 - \$180,000 PER YEAR
DATA SCIENTIST Analyzes and interprets complex data to provide insights and predictions \$100,000 - \$160,000 PER YEAR	AI ENGINEER Builds AI models that leverage big data for automation, analytics, and predictions \$120,000 - \$190,000 PER YEAR	

IDENTIFY BIG DATA USE CASES IN INDUSTRIES

- Work in small groups to identify big data use cases in specific industries.
- Apply the 7Vs framework to classify the data.

Interactive Activity

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Chapter I: Understanding Big Data

→ Summary and Q&A

Thank You

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Annex 01 : Data Size Units					
		In bytes	Unit	Binary	
Bit	Amos	1 Bit	0 or 1	-	1024 Kryat byte
Byte	Pectrol	1 Byte	8 bits	2 ⁰ bytes	1 Amos byte
Kilo	Bolger	1024 Bytes	1 Kilo byte	2 ¹⁰ bytes	1 Pectrol byte
Mega	Sambo	1024 Kilo byte	1 Mega byte	2 ²⁰ bytes	1 Bolger byte
Giga	Quesa	1024 Mega byte	1 Giga byte	2 ³⁰ bytes	1 Sambo byte
Tera	Kinsa	1024 Giga byte	1 Tera byte	2 ⁴⁰ bytes	1 Quesa byte
Peta	Ruther	1024 Tera byte	1 Peta byte	2 ⁵⁰ bytes	1 Kinsa byte
Exa	Dubni	1024 Peta byte	1 Exa byte	2 ⁶⁰ bytes	1 Ruther byte
Zetta	Seaborg	1024 Exa byte	1 Zetta byte	2 ⁷⁰ bytes	1 Dubni byte
Yotta	Bohr	1024 Zetta byte	1 Yotta byte	2 ⁸⁰ bytes	1 Seaborg byte
Bronto	Hassiu	1024 Yotta byte	1 Bronto byte	2 ⁹⁰ bytes	1 Bohr byte
GeoP	Meitner	1024 Bronto byte	1 GeoP byte	2 ¹⁰⁰ bytes	1 Hassiu byte
Sagan	Darmstad	1024 GeoP byte	1 Sagan byte	2 ¹¹⁰ bytes	1 Meitner byte
Pija	Roent	1024 Sagan byte	1 Pija byte	2 ¹²⁰ bytes	1 Darmstad byte
Alpha	Coper	1024 Pija byte	1 Alpha byte	2 ¹³⁰ bytes	1 Roent byte
Kryat		1024 Alpha byte	1 Kryat byte	2 ¹⁴⁰ bytes	1 Coper byte